

Tutorial 5: GNU Toolchain*Nanyang Technological University*

Q1 If we execute the following commands for the file, ce3103.c, what will be the respective output?
Explain the differences between the commands.

1. gcc ce3103.c
2. gcc -c ce3103.c

Q2 What is the command to use if we want to compile the file, ce3103.c such that the executable file is ce3103a that also support GDB debugging?

Q3 Suppose you are working on a C project, and you have a directory structure like this:

```
project/  
|  
|-- src/  
|   |-- main.c  
|   |-- utils.c  
|  
|-- include/  
|   |-- utils.h  
|   |-- common.h  
|  
|-- Makefile
```

You want to create a Makefile to build your project. The project contains two source files (main.c and utils.c) and uses header files (utils.h and common.h) located in the include directory.

Create a Makefile that defines the following targets and rules:

- myprogram target: Builds the executable myprogram.
- main.o and utils.o targets: Compile the respective source files into object files.
- Dependencies between the source files, object files, and header files.
- A clean target to remove generated files.

Explain the purpose of each part of your Makefile, including how dependencies are managed, how the clean target works, and how the Makefile ensures that the project is built correctly.

Provide step-by-step instructions on how to use your Makefile to build the project, specifying any necessary commands or options.